RESEARCH PROPOSAL

Computer Vision-Based Sri Lankan Food Recognition and Calorie Estimation System

Faculty of Information Technology

Horizon Campus

Full Name : G. Rashmi Dulashani

Student ID : ITBIN-2313-0031

Partner's Name : H.M. Imashi Dilshani

Module Name : Intelligent Systems

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1 RESEARCH GAP STATEMENT

1.1 Background and Existing Research

Artificial Intelligence (AI) and computer vision technologies have improved healthcare and nutrition monitoring systems in recent years. Deep learning models, especially Convolutional Neural Networks (CNNs), are widely used for automatic food recognition and classification. These systems are used in calorie-tracking applications, smart healthcare platforms, and dietary monitoring tools to support healthy lifestyles. Food recognition systems use techniques such as image classification, object detection, and image segmentation to identify food items from images.

Recent studies have shown that deep learning-based food recognition systems can achieve high classification accuracy when trained using large-scale food image datasets. Researchers commonly use transfer learning with pre-trained models such as ResNet, EfficientNet, and YOLO to improve performance while reducing training complexity. Benchmark datasets including Food-101, UECFOOD-256, and ChineseFoodNet are frequently used to evaluate these systems. Existing models have demonstrated strong performance on standardized food datasets containing visually clear and well-structured meals [1].

1.2 Current Techniques and Existing System Limitations

Most existing food-recognition systems rely on transfer learning approaches and pre-trained CNN architectures to classify food items and estimate calorie values. Some systems also combine object detection and portion-size estimation techniques to improve calorie calculation accuracy. These methods have significantly improved food classification performance and are widely applied in calorie-tracking applications and intelligent healthcare platforms.

However, most current research mainly focuses on Western and East Asian food datasets. Publicly available datasets usually contain simple meals with clearly separated food items, limited overlap, and standardized serving sizes. As a result, many existing models assume clear object boundaries and consistent nutritional information during classification and calorie estimation processes.

This creates a significant limitation when applying these systems to local South Asian food environments. Existing models are not designed to handle complex mixed meals containing multiple food items with similar textures, colors, and overlapping arrangements. Furthermore, most calorie estimation systems depend on Western nutritional databases and fixed serving assumptions, which may not accurately represent local dietary patterns [2].

This gap exists mainly because of the lack of publicly available Sri Lankan food image datasets, limited nutritional annotation for local meals, and the research community's stronger focus on international food datasets.

1.3 Identified Research Gap

Despite progress in AI-based food recognition, there are still major limitations when applying these systems to Sri Lankan food environments. Sri Lankan meals are highly complex compared to the structured meals commonly used in international datasets. A typical Sri Lankan plate often includes rice, multiple curries, sambols, vegetables, and side dishes served together, creating overlapping food items and unclear boundaries.

Food items such as kottu, string hoppers, hoppers, pittu, rice and curry, and pol sambol also show significant variation in appearance, ingredients, and preparation methods. These variations increase the difficulty of image classification and calorie estimation for models trained using international datasets.

In addition, existing calorie estimation systems are not well-suited for Sri Lankan foods because local meals vary greatly in portion size, oil usage, spices, and ingredient composition. The absence of a localized Sri Lankan food dataset and annotated nutritional database further limits accurate system development and evaluation [3].

Therefore, although existing food-recognition systems perform effectively on international benchmark datasets, their applicability to Sri Lankan mixed-food environments remains limited. No existing study has comprehensively evaluated whether transfer learning-based models trained on locally collected Sri Lankan food datasets can improve food-recognition and calorie-estimation accuracy for complex Sri Lankan meals.

Figure 1: Comparison between Structured Western Meals and Complex Sri Lankan Mixed-Food Meals

1.4 Consequence and Regional Relevance

The identified research gap is very important for Sri Lanka and the South Asian region in 2026. Because, Sri Lanka is facing an increase in lifestyle-related diseases such as diabetes, obesity, high blood pressure, and heart disease. At the same time, many people still have low awareness about nutrition tracking and calorie management compared to developed countries.

Most international calorie-tracking apps cannot correctly identify Sri Lankan foods. This reduces their usefulness for local users. When calorie values are wrong, people may follow incorrect diet plans, which can affect their health and nutrition monitoring [4], [5].

Because of this, there is a strong need for a local intelligent system that can recognize Sri Lankan foods and estimate calories more accurately.

2 RESEARCH QUESTION

2.1 Research Question

Does transfer learning-based deep learning using a locally collected Sri Lankan food image dataset achieve significantly higher food-recognition accuracy and calorie-estimation performance compared with models trained only on generic international food datasets for complex Sri Lankan mixed meals?

2.2 Falsifiability of the Research Question

This research question is falsifiable because experimental evaluation may show that using a locally collected Sri Lankan food dataset does not significantly improve food-recognition accuracy or calorie-estimation performance when compared with baseline models trained on international datasets.

3 SCOPE DEFINITION

3.1 Research Scope Category

This research falls under the category of applying an established AI technique to a new domain-specific dataset where it has not been widely studied before. The study applies transfer learning-based deep learning models for Sri Lankan food recognition and calorie estimation using locally collected food images. In addition, the research includes a comparative evaluation of existing CNN models under conditions specific to Sri Lankan mixed meals.

3.2 Scope of the Study

The study focuses on food image classification, calorie estimation, dataset preparation, transfer learning implementation, and model performance evaluation using Sri Lankan food images.

3.3 Out-of-Scope Areas

The research does not include real-time mobile application deployment, multilingual support, video-based food recognition, medical-grade dietary diagnosis, or advanced 3D portion-size reconstruction techniques. These areas are excluded to maintain a manageable research scope within the available time and computational resources [6].

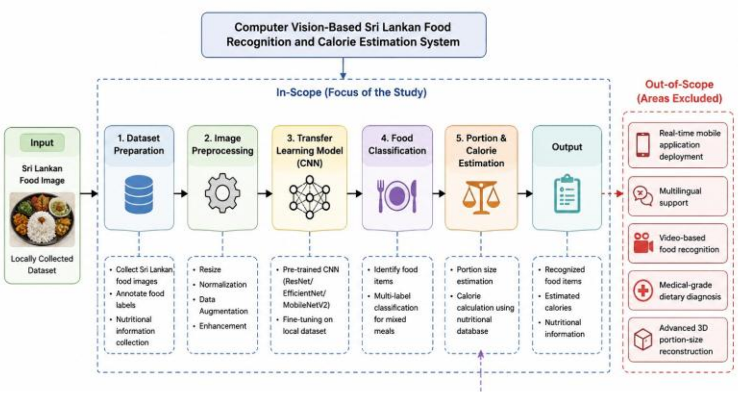

Figure 2: Scope of the Study

4 ANNOTATED BIBLIOGRAPHY

[1] M. Kalaimani, A. S. D, D. S. M, and S. P. Santhoshkumar, “Smart Food Recognition and Nutrition Estimation using Computer Vision,” in *2026 9th International Conference on Inventive Computation Technologies (ICICT)*, Kirtipur, Nepal: IEEE, Apr. 2026, pp. 1198–1204. doi: 10.1109/ICICT68280.2026.11510899.

Annotation:
This study proposed a computer vision-based food recognition and nutrition estimation system using deep learning techniques for intelligent dietary monitoring. The paper is relevant to the proposed research because it demonstrates how food recognition and nutritional estimation can be integrated within healthcare-related AI applications.

[2] H. Kagaya, K. Aizawa, and M. Ogawa, “Food Detection and Recognition Using Convolutional Neural Network,” in *Proceedings of the 22nd ACM international conference on Multimedia*, Orlando Florida USA: ACM, Nov. 2014, pp. 1085–1088. doi: 10.1145/2647868.2654970.

Annotation:
This paper introduced a CNN-based approach for food detection and recognition using image classification techniques. The study demonstrated the effectiveness of deep learning for food-recognition tasks and provides foundational knowledge for the proposed Sri Lankan food-recognition system.

[3] S. Yu, B. Feng, B. Zhu, and C. Wan, “A Real-life Chinese Dishes Recognition System Evolved from Full Training to Transfer Learning and Domain Adaptation,” in *Proceedings of the 2024 7th International Conference on Image and Graphics Processing*, Beijing China: ACM, Jan. 2024, pp. 135–139. doi: 10.1145/3647649.3647671.

Annotation:
This research developed a Chinese food-recognition system using transfer learning and domain adaptation techniques to improve classification performance on real-world food images. The paper is highly relevant because it addresses challenges related to local food diversity and transfer learning, which are similar to the proposed Sri Lankan food-recognition context.

[4] A. N. Yumang, D. E. S. Banguilan, and C. K. S. Veneracion, “Fruit Calories Estimation using Convolutional Neural Network,” in *Proceedings of the 2023 13th International Conference on Biomedical Engineering and Technology*, Tokyo Japan: ACM, Jun. 2023, pp. 231–235. doi: 10.1145/3620679.3620720.

Annotation:
This study proposed a CNN-based calorie estimation system for fruit images using image-processing techniques. The research contributes to understanding how computer vision can

support calorie estimation tasks, which directly relates to the calorie-estimation component of the proposed study.

[5] H. Qi, B. Zhu, C.-W. Ngo, J. Chen, and E.-P. Lim, “Advancing Food Nutrition Estimation via Visual-Ingredient Feature Fusion,” in *Proceedings of the 2025 International Conference on Multimedia Retrieval*, Chicago IL USA: ACM, Jun. 2025, pp. 1091–1099. doi: 10.1145/3731715.3733269.

Annotation:
This paper introduced a feature-fusion approach combining visual and ingredient-based information for food nutrition estimation. The study is relevant because it highlights advanced methods for improving nutritional estimation accuracy in complex food-recognition systems.

[6] J. Patel and K. Modi, “Indian Food Image Classification and Recognition with Transfer Learning Technique Using MobileNetV3 and Data Augmentation,” in *The 4th International Electronic Conference on Applied Sciences*, MDPI, Oct. 2023, p. 197. doi: 10.3390/ASEC2023-15341.

Annotation:
This study applied transfer learning and data augmentation techniques for Indian food image classification using MobileNetV3 architecture. The paper is important to the proposed research because Indian and Sri Lankan foods share similar mixed-meal characteristics and visual complexity.

[7] J. A. Patel, D. L. Labana, G. V. Lakhani, and R. K. Vaghela, “Deep Learning and Transfer Learning Models for Indian Food Classification,” in *The 6th International Electronic Conference on Applied Sciences*, MDPI, Feb. 2026, p. 14. doi: 10.3390/engproc2026124014.

Annotation:
This paper compared deep learning and transfer learning models for Indian food classification tasks. The study supports the proposed research by providing comparative insights into model performance for South Asian food-recognition environments.

[8] S. Yu, B. Feng, B. Zhu, and C. Wan, “A Real-life Chinese Dishes Recognition System Evolved from Full Training to Transfer Learning and Domain Adaptation,” in *Proceedings of the 2024 7th International Conference on Image and Graphics Processing*, Beijing China: ACM, Jan. 2024, pp. 135–139. doi: 10.1145/3647649.3647671.

Annotation:
This study explored domain adaptation and transfer learning methods for recognizing complex Chinese dishes in real-world conditions. The research is relevant because it

demonstrates techniques that may improve recognition performance for Sri Lankan mixed-food datasets.

[9] R. Abiyev and J. Adepoju, “Automatic Food Recognition Using Deep Convolutional Neural Networks with Self-attention Mechanism,” *Hum.-Centric Intell. Syst.*, vol. 4, no. 1, pp. 171–186, Jan. 2024, doi: 10.1007/s44230-023-00057-9.

Annotation:
This paper proposed a food-recognition model combining CNNs with self-attention mechanisms to improve classification accuracy. The study informs the proposed research by presenting advanced deep learning techniques suitable for handling visually complex food images.

[10] B. N. Jagadesh *et al.*, “Enhancing Food Recognition Accuracy using Hybrid Transformer Models and Image Preprocessing Techniques,” *Sci. Rep.*, vol. 15, no. 1, p. 5591, Feb. 2025, doi: 10.1038/s41598-025-90244-4.

Annotation:
This research introduced hybrid transformer models and image preprocessing methods to improve food-recognition accuracy. The paper is relevant because it demonstrates how advanced architectures can improve performance for challenging food image datasets.

[11] D. Liu, E. Zuo, D. Wang, L. He, L. Dong, and X. Lu, “Deep Learning in Food Image Recognition: A Comprehensive Review,” *Appl. Sci.*, vol. 15, no. 14, p. 7626, Jul. 2025, doi: 10.3390/app15147626.

Annotation:
This review paper analyzed recent deep learning techniques, datasets, and challenges in food image recognition systems. The study strongly supports the proposed research by identifying limitations related to dataset diversity, mixed-food recognition, and calorie-estimation accuracy.

[12] Venkata Ravi Kiran Kolla, “A Novel Machine Learning Approach to Calorie Estimation for Diabetes and Obesity Management,” *Int. J. Innov. Eng. Res. Technol.*, vol. 11, no. 12, pp. 8–16, Dec. 2024, doi: 10.26662/ijiert.v11i12.pp8-16.

Annotation:
This paper proposed a machine learning-based calorie estimation system designed to support diabetes and obesity management. The study is relevant because it highlights the healthcare importance of accurate calorie estimation systems for preventive health monitoring and nutritional management.

5 TARGET JOURNAL

5.1 Selected Journal

The selected target journal for this proposed research is IEEE Access.

5.2 Scope Fit

According to the journal's aims and Scope statement, IEEE Access publishes multidisciplinary research related to artificial intelligence, machine learning, computer vision, healthcare informatics, and intelligent systems. The proposed study aligns with this scope because it applies deep learning and computer vision techniques for Sri Lankan food recognition and calorie estimation within a healthcare-related context. The research also focuses on transfer learning, image classification, and intelligent nutritional monitoring systems, which are suitable topics for the journal [8].

5.3 Impact Factor and Tier

IEEE Access is a well-recognized Q1 journal with a strong impact factor and broad international visibility. It is considered an appropriate target for a first research publication because the journal accepts practical AI-based applications supported by experimental evaluation and comparative analysis. The scope and complexity of the proposed study are suitable for this publication level.

5.4 Open-Access Policy and APC

IEEE Access follows a fully open-access publishing model and requires an Article Processing Charge (APC) for accepted papers. However, IEEE provides waiver and discount opportunities for researchers from developing countries, including Sri Lanka. This makes the journal a realistic and accessible publication target for the proposed research.

6 REFERENCES

[1] M. Kalaimani, A. S. D, D. S. M, and S. P. Santhoshkumar, "Smart Food Recognition and Nutrition Estimation using Computer Vision," in *2026 9th International Conference on Inventive Computation Technologies (ICICT)*, Kirtipur, Nepal: IEEE, Apr. 2026, pp. 1198–1204. doi: 10.1109/ICICT68280.2026.11510899.

[2] "Food Recognition and Calorie Estimation using Machine Learning | Ieee Xpert ,Ieee Xpert, ieee projects final year cse students." Accessed: May 22, 2026. [Online]. Available: https://www.ieeexpert.com/python-projects/food-recognition-and-calorie-estimation-using-machine-learning/

[3] "SriLankan Foods Model Overview." Accessed: May 22, 2026. [Online]. Available: https://universe.roboflow.com/calorieesimate/srilankanfoods

[4] H. Kagaya, K. Aizawa, and M. Ogawa, "Food Detection and Recognition Using Convolutional Neural Network," in *Proceedings of the 22nd ACM international conference on Multimedia*, Orlando Florida USA: ACM, Nov. 2014, pp. 1085–1088. doi: 10.1145/2647868.2654970.

[5] H. Qi, B. Zhu, C.-W. Ngo, J. Chen, and E.-P. Lim, "Advancing Food Nutrition Estimation via Visual-Ingredient Feature Fusion," in *Proceedings of the 2025 International Conference on Multimedia Retrieval*, Chicago IL USA: ACM, Jun. 2025, pp. 1091–1099. doi: 10.1145/3731715.3733269.

[6] "Food Recognition Model with Deep Learning Techniques [Updated]." Accessed: May 22, 2026. [Online]. Available: https://www.labellerr.com/blog/food-recognition-and-classification-using-deep-learning/

[7] Venkata Ravi Kiran Kolla, "A Novel Machine Learning Approach to Calorie Estimation for Diabetes and Obesity Management," *Int. J. Innov. Eng. Res. Technol.*, vol. 11, no. 12, pp. 8–16, Dec. 2024, doi: 10.26662/ijiert.v11i12.pp8-16.

[8] "IEEE Access." Accessed: May 22, 2026. [Online]. Available: https://ieeeaccess.ieee.org/

7 ACADEMIC INTEGRITY DECLARATION

IT41043 - Intelligent Systems | Assignment 1: Research Proposal

I declare that this proposal is my own original work. I confirm that:

1. I have not used any large language model (LLM), generative AI tool, or AI writing assistant to draft, outline, paraphrase, or edit any portion of this document.
2. I have personally read every source included in my annotated bibliography.
3. All sources are correctly attributed. I have not fabricated, misrepresented, or plagiarised any reference.
4. I understand that submitting work that is not my own constitutes academic misconduct and may result in a zero mark and disciplinary action under the Horizon Campus Academic Integrity Policy.

Full Name: G. Rashmi Dulashani

Student ID: ITBIN-2313-0031

Partner’s Name: H.M. Imashi Dilshani

Date: 21/05/2026

Signature: